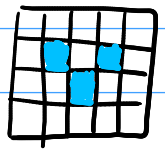
MATE 664 L12

Intermission 1

Comment on interactive demo

(arbitrary math model for phase transformation)



Grid-based "Boltzmann machine"

each grid can be either $\begin{cases} \text{occupied (blue)} \\ \text{unoccupied (blank)} \end{cases}$

Total energy in this system

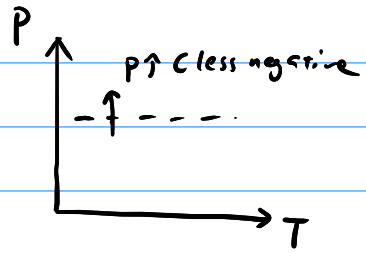
$$G = U + PV - TS$$

Simluation in this system

$U \Rightarrow E$ (internal energy)

$$C = \frac{PV - TS}{N} \quad \text{(modified}$$

From here $E_{total} = E - C \cdot N$



$P\uparrow$ less negative C

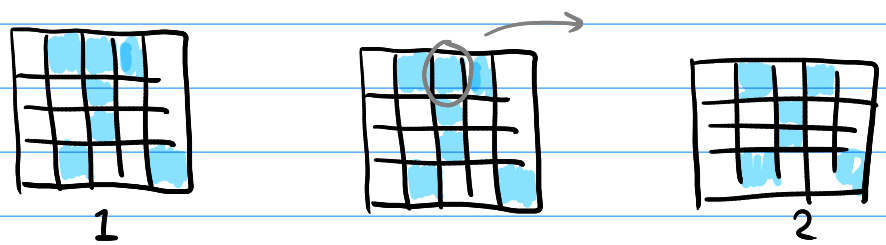
S from gas → liquid $S\downarrow$ less negative

Why is the equibrium?

Color is the total number of maters

How does simulation work?

old system $\xrightarrow{\text{pick a grid}}$ probability of grid become the opposite?



$$P(1) \propto \exp\left(-\frac{E_{N_1} + C \cdot N_1}{T}\right) \qquad P(2) = \exp\left(-\frac{E_{N_2} + C \cdot N_2}{T}\right)$$